Day-12 Task-3: Schema design and querying on E-commerce dataset (Day-11 Task-3 dataset)

1. Database creation

```
CREATE DATABASE DB_DAY_12_TASK_3;
USE DB_DAY_12_TASK_3;
```

2. Schema creation (table)

```
CREATE TABLE CustomerOrders(
      order_id VARCHAR(16) PRIMARY KEY,
    order_datetime TIMESTAMP NOT NULL,
    customer_id VARCHAR(16) NOT NULL,
    city VARCHAR(32),
    product_name VARCHAR(64) NOT NULL,
    category VARCHAR(32),
    cost_per_unit DECIMAL NOT NULL CHECK (cost_per_unit > 0),
    quantity INTEGER NOT NULL CHECK (quantity >= 0),
    payment_method VARCHAR(24),
    revenue DECIMAL NOT NULL CHECK (revenue >= 0)
);
```

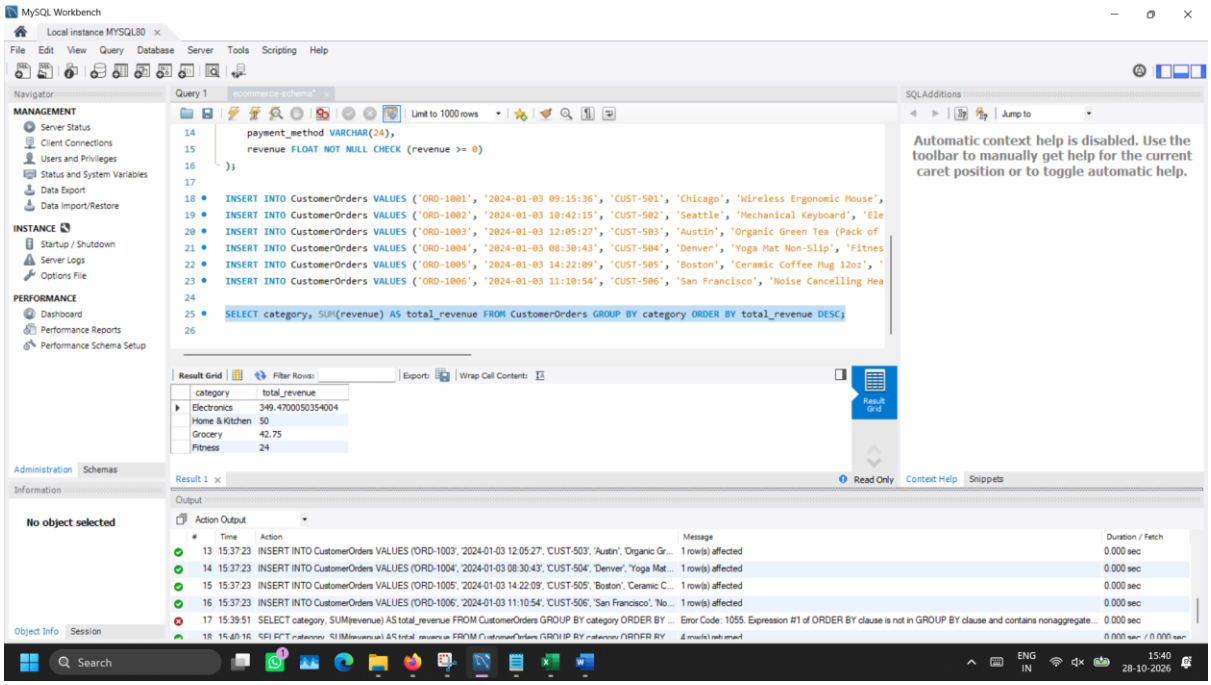
3. Data Insertion

```
INSERT INTO CustomerOrders VALUES ('ORD-1001', '2024-01-03 09:15:36',
'CUST-501', 'Chicago', 'Wireless Ergonomic Mouse', 'Electronics', 29.99,
2, 'Credit Card', 59.98);
INSERT INTO CustomerOrders VALUES ('ORD-1002', '2024-01-03 10:42:15',
'CUST-502', 'Seattle', 'Mechanical Keyboard', 'Electronics', 89.5, 1,
'PayPal', 89.5);
INSERT INTO CustomerOrders VALUES ('ORD-1003', '2024-01-03 12:05:27',
'CUST-503', 'Austin', 'Organic Green Tea (Pack of 50)', 'Grocery', 14.25,
3, 'Debit Card', 42.75);
INSERT INTO CustomerOrders VALUES ('ORD-1004', '2024-01-03 08:30:43',
'CUST-504', 'Denver', 'Yoga Mat Non-Slip', 'Fitness', 24, 1, 'Credit
Card', 24);
INSERT INTO CustomerOrders VALUES ('ORD-1005', '2024-01-03 14:22:09',
'CUST-505', 'Boston', 'Ceramic Coffee Mug 12oz', 'Home & Kitchen', 12.5,
4, 'Apple Pay', 50);
INSERT INTO CustomerOrders VALUES ('ORD-1006', '2024-01-03 11:10:54',
'CUST-506', 'San Francisco', 'Noise Cancelling Headphones', 'Electronics',
199.99, 1, 'Credit Card', 199.99);
```

4. Grouping, Sorting and applying aggregate function

Display total revenue of each category of products in descending order

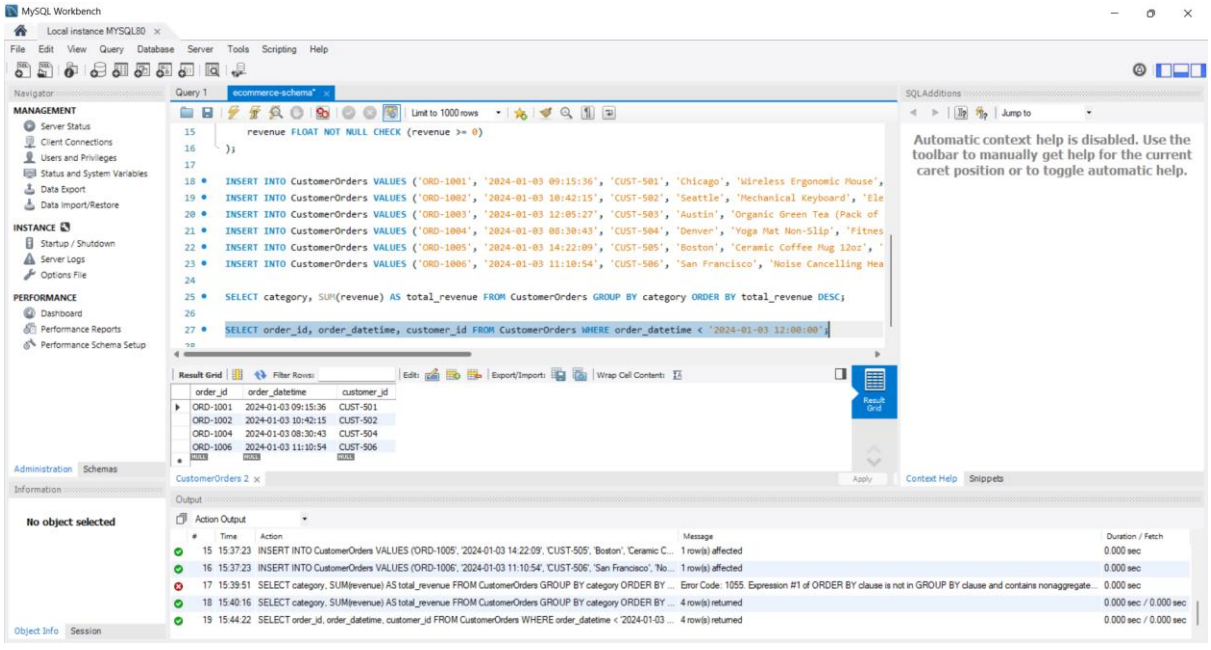
```
SELECT category, SUM(revenue) AS total_revenue FROM CustomerOrders GROUP BY category ORDER BY total_revenue DESC;
```



5. WHERE clause usage

Retrieve orders info which was made before noon of January 03, 2024

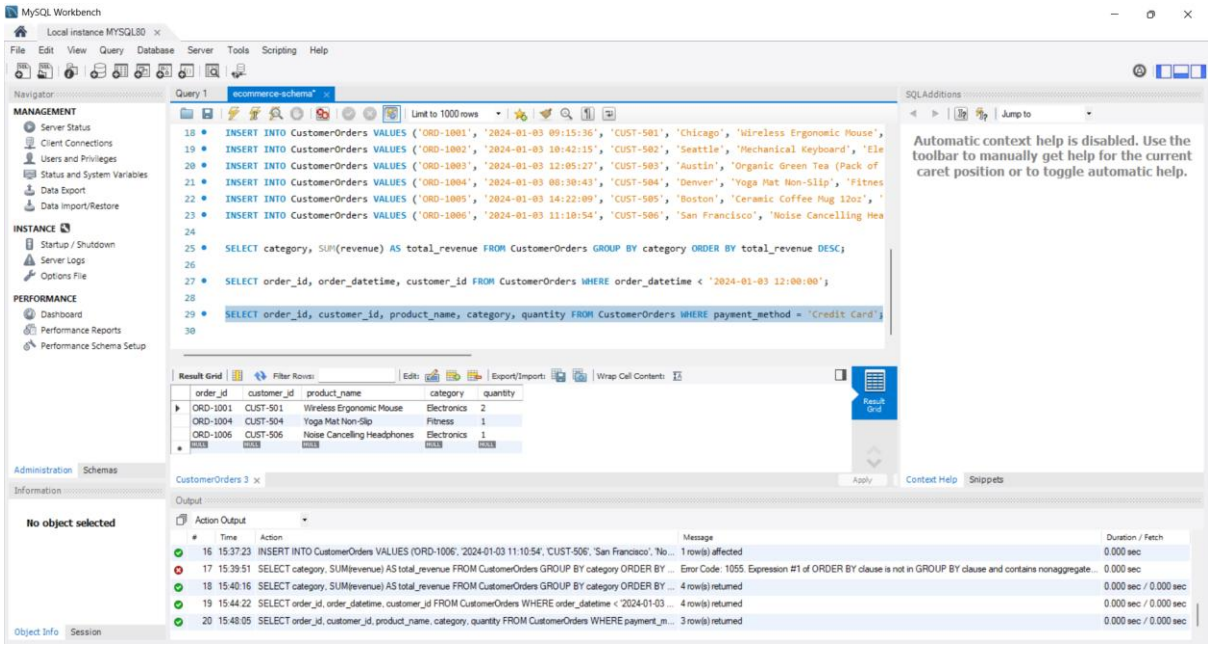
```
SELECT order_id, order_datetime, customer_id FROM CustomerOrders WHERE order_datetime < '2024-01-03 12:00:00';
```



6. Applying Filter

Identify orders whose payment is made only through Credit Card.

```
SELECT order_id, customer_id, product_name, category, quantity FROM
CustomerOrders WHERE payment_method = 'Credit Card';
```



7. Grouping, average and identifying top 2 categories

Find top 2 product categories with highest average revenue.

```
SELECT category, AVG(revenue) as average_revenue FROM CustomerOrders GROUP
BY category ORDER BY average_revenue DESC LIMIT 2;
```

